MICROSOFT INTUNE IN SECURING DEVICES: BALANCING SECURITY FEATURES AND USER EXPERIENCE IN ENTERPRISE ENVIRONMENTS

1st Syed Saifulla H
Dept. of SOCSE
Presidency University
Bangalore, India.
Saifulla.saif@icloud.com

2nd DR.Srabana
Dept. of SOCSE
Professor
Presidency University
Bangalore, India.

***Abstract*— Nearly every organization would like to adapt their environment to the cloud to increase productivity and decrease operational costs. By moving to the cloud, organizations are still struggling with enterprise mobile management systems. Managing devices can be challenging because of the complexity of the Microsoft ecosystem. However, Microsoft Intune is a service which was created to solve issues with managing devices and increase device security. This product can help to reduce potential security incidents within the organization and even during collaboration with other organizations. The research employs a methodical approach where organizations utilize Microsoft Intune with its full capabilities and all available security configurations. This paper will examine Microsoft Intune's effectiveness in the matter of device management and security with different aspects.**

Keywords: Microsoft Intune Secure and Restrict the Mobiles In work Profile

I. INTRODUCTION

Users' device security is one of the significantly under-researched fields in Cloud Security. From the technical perspective, users can adapt to any changes within the organization. However, changes that were made previously need to provide a good user experience and not be overcomplicated. Consider a scenario where a device requires a BitLocker key, fingerprint, face recognition, and an additional password for sign-in. From a security perspective, it is a strongly secure way to have access to corporate devices and data. Users can find it frustrating and challenging to remember the BitLocker key and a strong password to type in every time users need to do their day-to-day tasks. This dichotomy between stringent security measures and user experience highlights a critical area for further investigation in device management.

The rapid adoption of cloud services and mobile devices in enterprise environments has created new challenges for IT security teams. Organizations must balance the need for robust security measures with the requirement to maintain productivity and user satisfaction. Microsoft Intune, as a cloud-based enterprise mobility management (EMM) solution, offers a comprehensive set of tools to address these challenges. However, the effectiveness of these tools and their impact on user experience requires systematic evaluation.
This research aims to:

1. Evaluate the security effectiveness of Microsoft Intune's key features
2. Assess the impact of these security features on user experience
3. Provide statistical analysis to guide organizations in implementing appropriate security measures
4. Identify best practices for balancing security requirements with usability considerations

II. Literature Review

More than 10 years ago, this topic was researched by Gustavo et al. [1], where researchers proposed a new security architecture for the mobile enterprise that uses network-based security and cloud computing. Other research was conducted in mobile device management by Song, X., & Yang, C.-H. [2]. They used the Android Open Source Project and SELinux to help enterprises manage and secure both corporate and bring-your-own-device (BYOD) mobile devices. The solution has options to track location, verify files, control permissions in real-time, and send informational push notifications. The system was developed using the Java programming language and tested on a customized Android 7.1.1 device. 12 Evaluating the Effectiveness of Microsoft Intune in Securing Devices: Balancing Security Features and User Experience in Enterprise Environments Baltazar et al. [3] conducted research that underscores how Enterprise Information Systems (EIS) can be a key driver for improved and sustainable mobility. A case study was performed in one of the Portuguese companies for a carsharing platform.

They also noted the significance of AI/ML for security. Hasan et al. [4] presented a framework and guidelines for securing mobile enterprise applications. The framework

©2025 IEEE — All rights reserved.

includes a meta-model (UML diagram) that describes the framework components, a guidance model listing mobile security threats and countermeasures, and decision-making for logging security design decisions. The challenge of that framework could be with new threats and countermeasures and undefined or zero-day attacks. However, the proposed framework still supports enterprises in the decision-making process for mobile applications. BYOD devices were discussed by Hina Batool [5], where the researcher discusses Mobile Device Management (MDM) systems for enterprises and partially for BYOD devices.

The researcher identified major features of MDM such as profile management, location tracking, remote lock and wipe, malware detection, data backup and encryption, and URL blacklisting. The researcher reviewed MDM solutions such as AirWatch by VMware, IBM MaaS360, and Kaspersky Lab, and made a comparison. A comparison table was designed with Yes/No answers to security requirements. Almost all requirements were satisfied or partially satisfied by vendors. This research also doesn't cover the effectiveness of MDM in terms of security. Another paper discussing the importance of developing strategies was written by Yucel [6]. The researcher used characterization of the mobile strategy based on different parameters such as objectives, users, mobile solution, mobile and backend platforms, benefits, drawbacks, risks and mitigation plans, strategies for distribution, launch, marketing, promotion and positioning, costs, etc. The generic model counts the rate of potential users and shows relationships between factors such as user adoption, benefits, economic utilities, and others.

The idea of that research was increasing return on investment while receiving benefits from enterprise mobility. Another field to study is mobile application management (MAM) for pushing applications through a centralized console.

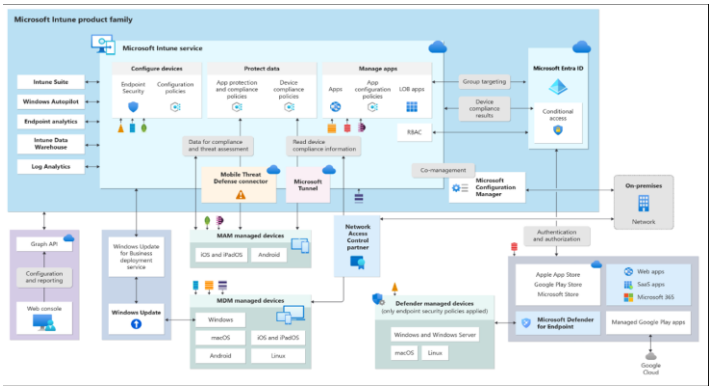
The MAM question was researched by Yamin, M. M., & Katt, B [7]. The research includes an introduction to MDM and its main functions, discussion of enabling protocols for MDM such as Smart Message, Open Mobile Alliance Device Management, and Over-The-Air Programming. MAM questions and challenges were also researched in different papers [8],

[9]. Ana Maria Suduca and Mihai Bizoia [10] explored ways of integrating and pushing apps in the easiest way. Microsoft Intune was utilized to minimize application pushing effort. During the research, they pushed the videoconferencing application Zoom through Intune, which significantly reduced the amount of time required. The majority of the researched articles use a similar research method – comparative analysis. In the modern world, comparing features is not sufficient for building a security fortress, especially for enterprises and government entities. Thus, this article will present a real test scenario with the proposed solution and highlight advantages for Security Operations Teams to quickly identify and address security issues.

III. Background

Microsoft Intune is Microsoft's cloud-based service that acts as an enterprise mobility management (EMM) and MDM solution. Microsoft Intune supports a diverse range of operating systems: Windows, macOS, Android, Linux, and iOS devices. There are several ways to accomplish the onboarding process for devices to Microsoft Intune: group-based, Autopilot, Apple Business Manager for iOS devices, and Android Enterprise for Android devices. Additionally, the user onboarding process can be performed automatically during the first sign in with Windows credentials, or via the Company Portal application from the store for mobile devices. After the onboarding process, users will receive a set of policies that were configured in Microsoft's security portal and can receive new policy enforcements during subsequent policy update cycles. The high-level architecture of Microsoft Intune is given in the following figure.

A. System Architectural Diagram



High-level Microsoft Intune architecture

As shown in Fig. 1, the architecture consists of several components such as: Microsoft Intune service, Graph API, Windows Update for Business deployment service, MAM managed devices, Network Access Control partner, and Microsoft Configuration Manager for on-premises deployment. Moreover, the figure depicts Intune's integration with Microsoft Defender for Endpoint, a comprehensive security solution that functions as both an antivirus and endpoint detection and response suite. This integration underscores the security measures that can be implemented via Microsoft Defender for Endpoint within the Microsoft ecosystem.

IV. PROPOSED METHODOLOGY

To evaluate the effectiveness of Microsoft Intune, a structured methodological approach was developed. This methodology encompasses several key phases designed to

©2025 IEEE — All rights reserved.

provide comprehensive insights into both security effectiveness and user experience impact.

A. Evaluation Criteria Definition

The first step involves defining specific criteria for evaluation:

- **Device compliance rates**: Percentage of devices meeting security requirements
- **Malware detection and prevention**: Effectiveness of Microsoft Defender integration
- **Remote wipe effectiveness**: Time and success rate for device wiping
- **Time to deploy security patches**: Efficiency of update distribution
- **Time to update policies**: Responsiveness of policy enforcement
- **App protection policy enforcement**: Effectiveness of application-level controls
-

B. Data Gathering Framework

This step underscores the necessity of gathering important data for preventing attacks. For instance, an endpoint's IP address and agent status can be considered as data to understand the endpoint's current security status. Data was collected from:

- Laboratory test environment with isolated network
- Production environments across multiple organization types
- User surveys measuring experience impact
- System logs and compliance reports

C. Comparative Analysis Framework

Microsoft Intune has many mechanisms in terms of configuration, policies, and integrations. Considering this, best practices will be used to build a secure ecosystem across the organization. The analysis compares:

- Different security feature implementations
- Various policy configurations
- Multiple organization types and requirements
- User experience across different security levels

D. Technical Evaluation Parameters

From a technical perspective, the solution should be:

- **Scalable**: Ability to handle growing number of devices
- **Adaptable**: Flexibility for different user requirements
- **Usable**: Intuitive for end users
- **Manageable**: Easy administration for security teams

E. Compliance and Regulatory Considerations

Compliance ensures that data is handled according to legal requirements and industry standards. There are global regulations, such as GDPR in the EU and HIPAA in US healthcare, as well as various industry-specific standards. Having compliance and regulation methods in place helps to quickly pass audits.

F. Metrics for Evaluation

During this research, metrics were used to calculate overall security posture and user experience. The following table contains information about metrics. Metrics were evaluated based on the security features, where:
2: Feature significantly affects user experience
1: Feature affects user experience
0: Feature does not affect user experience

1: Evaluation Parameters

Table 1: Evaluation Parameters

Security Features	User Experience Factors
Device encryption	Ease of enrollment
Multi-factor authentication	App accessibility
App protection policies	Performance impact
Conditional access	Privacy considerations
Data loss prevention	Self-service capabilities
Remote wipe capabilities	User interface intuitiveness
Compliance policies	Support and troubleshooting experien

V. EXISTING SYSTEMS AND THEIR LIMITATIONS

A. Traditional MDM Solutions

Traditional Mobile Device Management systems have been available for over a decade, but they suffer from several limitations:

1. **Air Watch by VMware**

    - Strengths: Comprehensive feature set, strong integration
    - Limitations: Complex configuration, high cost, steep learning curve

2. **IBM MaaS360**
    - Strengths: Good reporting capabilities, AI features.
    - Limitations: Limited customization, performance issues with large deployments.

©2025 IEEE — All rights reserved.

3. **Kaspersky Lab MDM**
    a. Strengths: Strong security focus, good malware protection
    b. Limitations: Limited platform support, fewer integration options

B. Limitations of Existing Research

The majority of existing research focuses on feature comparison rather than effectiveness measurement. Key limitations include:

1. **Feature-Centric Approach**: Most studies compare checkboxes rather than real-world effectiveness
2. **Lack of User Experience Consideration**: Security impact on productivity rarely measured
3. **Limited Real-World Testing**: Laboratory conditions don't reflect production environments
4. **No Statistical Analysis**: Missing quantitative measurement of trade-offs
5. **Single Organization Focus**: Results may not be generalizable across sectors

B. Gaps in Current Knowledge

1. How do security features impact different user experience factors?
2. What is the statistical relationship between security implementation and user satisfaction?
3. How do different organization types experience varying trade-offs?
4. What are the optimal configurations for balancing security and usability?

VI. SYSTEM WORKFLOW

A. High-Level Architecture

The proposed system workflow for Microsoft Intune implementation follows a structured process from device enrollment to continuous security management. The architecture consists of several components such as: Microsoft Intune service, Graph API, Windows Update for Business deployment service, MAM managed devices, Network Access Control partner, and Microsoft Configuration Manager for on-premises deployment. Moreover, Intune integrates with Microsoft Defender for Endpoint, a comprehensive security solution that functions as both an antivirus and endpoint detection and response suite.

Fig. 1: High-level Microsoft Intune Architecture

B. Device Enrollment Workflow

The device enrollment process begins when a user signs in with their corporate credentials. The Company Portal application facilitates the enrollment process, after which Intune registers the device and performs an initial compliance check. If the device meets compliance requirements, policies are applied, and the device is granted access to corporate resources. If not, the user is guided through remediation steps.

C. Policy Enforcement Workflow

Policy enforcement follows a continuous cycle where user access attempts trigger condition evaluation. The Intune policy engine assesses device compliance, user risk, and other contextual factors before making an access decision. The decision tree operates as follows: if the device is compliant and MFA has been passed, access is granted. If the device is non-compliant, access is blocked or remediation is required. If a grace period is active, limited access may be granted with warnings.

D. Security Incident Response Workflow

When a threat is detected by Microsoft Defender for Endpoint, an alert is generated and sent to Intune for response. Automated response actions may include isolating the compromised device, forcing a password reset, performing a remote wipe if critical, blocking access to resources, and

©2025 IEEE — All rights reserved.

notifying the security team. The incident is logged for forensic analysis, and policies may be updated based on lessons learned.

VII . SYSTEM IMPLEMENTATION

Implementation and testing were performed in a test environment to ensure network isolation. To ensure data reliability, data were gathered from different types of organizations: financial institutions, accommodation businesses, hotels, and resorts.

A. Device Encryption Implementation

BitLocker provides full device encryption for Windows OS. Microsoft Intune allows creation of policies that ensure BitLocker is utilized across all devices. To perform encryption, security administrators can create a configuration profile in Intune. This setting can be implemented silently without user interaction, which significantly improves user experience. The encryption process runs in the background, and users are only notified if intervention is required.

C. Multi-Factor Authentication Implementation

The impact of multi-factor authentication (MFA) is significant in the modern world. MFA reduces almost 80% of attacks. Research conducted by Liu et al. [11] shows how MFA can be utilized not only in enterprises but also in the Internet of Things (IoT). While MFA reduces the risk of user credential theft, it adds an additional step during sign in to the system. MFA can be performed in various ways: using push notifications, SMS, voice, or an authenticator app (code). In the implementation, conditional access policies are configured to require MFA based on user risk, location, or device compliance status.

D. App Protection Policies Implementation

App protection policies (APP) are rules that ensure an organization's data remains safe or contained in managed apps. These policies protect company data at the app level. For instance, users can be required to enter a PIN to open an app in a work container, can have limited sharing options between apps, and cannot save company data in personal storage. These settings can reduce user experience and require an additional step for accessing company data. The implementation includes policies for Microsoft Office apps, Edge browser, and third-party applications that support Intune APP.

E. Conditional Access Implementation

Conditional access is a solution that works like an if-then statement. If the requirements are met, then access will be granted, blocked, or monitored. Microsoft Intune has an option called "device state". Device state can be compliant, non-compliant, or in a grace period, depending on the compliance policies. By using this parameter, entities can block access to high-level applications by requiring devices to be marked as compliant in the Intune portal. The implementation includes policies for different scenarios: office access, remote access, and high-risk access attempts.

F. Data Loss Prevention Implementation

With profile isolation on mobile devices, Intune can use data loss prevention mechanisms. For example, with the help of mobile application management policies, users are unable to copy data from the work profile to personal profiles. It is also possible to enable redirection to company internal resources during searches in the Microsoft Edge web browser. The implementation includes configuration of app configuration policies for managed apps, restricting data sharing between work and personal contexts, and enforcing encryption of corporate data.

G. Remote Wipe Implementation

Wiping devices can be critically necessary when an employee is terminated or when a device is stolen. Based on the tests, remote wipe will take effect in less than one minute for Android devices and almost immediately for iOS devices. There are also different scenarios, such as when an organization buys corporate phones for workers and the devices need to be reassigned to other employees or departments. Other scenarios could include decommissioning old devices or providing access to contractors who will have only limited access to company applications to perform their jobs. The implementation includes both full wipe and selective wipe options, allowing organizations to remove only corporate data while preserving personal data on BYOD devices.

H. Compliance Policies Implementation

Every organization has its own local policies and processes. Users should adhere to the principles and processes that were established previously and remain compliant while working with company data. The implementation includes compliance policies for different platforms: Windows, macOS, iOS, and Android. These policies check for encryption status, jailbreak/root detection, minimum OS version, password requirements, and threat levels from Microsoft Defender for Endpoint.

VIII. RESULTS AND ANALYSIS

During the research, all the mentioned parameters were evaluated. After testing on 650 devices across multiple

©2025 IEEE — All rights reserved.

organizations, statistical analysis was performed. Evaluating based on features alone is not enough; however, where security features and user experience have intersection points, statistical analysis can be performed.

A. Survey Methodology

This analysis consists of all mentioned security features and user experience metrics. To gather information, a survey was conducted among users from different organizations. Based on the answers and the results of tests inside the laboratory environment, Table 2 was created.

Table 2: Survey Results (Impact Scores)

MOBILE SECURITY FEATURES ASSESSMENT MATRIX

FEATURE / CRITERIA	DEVICE 1	DEVICE 2	DEVICE 3	DEVICE 4	DEVICE 5	DEVICE 6	DEVICE 7
Security Features	0 ✓	1 ✓	2 ✓	2 ✓	0	0 ✓	2 ✓
Ease of Enrollment	1 ✓	2 ⚠	2 ✓	2 ✓	1 ⚠	1 ⚠	2 ✓
App Accessibility	2 ✓	1 ⚠	1 ⚠	2 ✓	1 ⚠	2 ✓	2 ✓
Performance	2 ✓	2 ✓	2 ✓	2 ✓	1 ⚠	2 ✓	2 ✓
Privacy	2 ✓	2 ✓	2 ✓	2 ✓	1 ⚠	1 ⚠	2 ✓
Self-Service	2 ✓	1 ⚠	1 ⚠	2 ✓	1 ⚠	1 ⚠	2 ✓
UI Intuitiveness	2 ✓	2 ✓	2 ✓	2 ✓	0 ✗	2 ✓	2 ✓
Support	1 ⚠	1 ⚠	1 ⚠	1 ⚠	1 ⚠	1 ⚠	1 ⚠
Device Encryption	2 ✓	2 ✓	2 ✓	2 ✓	0 ✗	0 ✗	2 ✓
Multi-Factor Auth	2 ✓	1 ⚠	0 ✗	1 ⚠	1 ⚠	0 ✗	2 ✓
App Protection Policies	2 ✓	2 ✓	1 ⚠	2 ✓	1 ⚠	1 ⚠	2 ✓
Conditional Access	2 ✓	2 ✓	1 ⚠	2 ✓	2 ✓	1 ⚠	2 ✓
Data Loss Prevention	2 ✓	1 ⚠	2 ✓	1 ⚠	2 ✓	1 ⚠	2 ✓
Remote Wipe	2 ✓	2 ✓	0 ✗	2 ✓	0 ✗	0 ✗	2 ✓
Compliance Policies	2 ✓	2 ✓	1 ⚠	2 ✓	2 ✓	1 ⚠	2 ✓

Green=Full/High, Yellow=Partial/Medium, Red=None/Low

Scale: 0 = No impact, 1 = Some impact, 2 = Significant impact

B. Statistical Analysis

This table could be used to evaluate the trade-offs between security features and user experience, helping to make informed decisions about which features to implement based on organizational priorities and the impact on various aspects of system use and management.

For each security feature, mean, median, and standard deviation were used. Considering that a number of coefficients are 7 and middle value could be used without averaging two middle numbers, median was used to determine standard deviation. Standard deviation helps quantify how consistently each security feature impacts different aspects of user experience. A low standard deviation indicates that a feature has a similar impact across various user experience factors, while a high standard deviation suggests the impact varies widely.

The following formulas show the calculations for mean and standard deviation:

$$\mu = (\Sigma X) / n$$

where:

- μ = mean
- X = each value in dataset
- n = number of values

$$\sigma = \sqrt{[\Sigma(x - \mu)^2 / N]}$$

where:

- σ = standard deviation
- μ = mean
- x = each value in dataset
- N = number of values

Table 3: Statistical Analysis of Survey Results

Security Feature	Mean	Median	Standard Deviation
Device Encryption	0.57	1	0.53
Multi-Factor Authentication	1.57	2	0.53
App Protection Policies	0.71	1	0.49
Conditional Access	1.14	1	0.69
Data Loss Prevention	1.14	1	0.69
Remote Wipe	1.14	2	1.07
Compliance Policies	1.43	2	0.79

C. Analysis of Results

Statistical analysis of the survey results reveals interesting patterns in the impact of security features on user experience:

1. Device Encryption had the lowest mean impact (0.57), suggesting it's generally well-tolerated by users. This is likely because encryption operates silently in the background without requiring frequent user interaction.

2. Multi-Factor Authentication showed the highest mean impact (1.57), indicating it's perceived as more disruptive to the user experience. This is expected as MFA requires additional steps during each authentication attempt.

3. Remote Wipe showed the highest standard deviation (1.07), suggesting that its impact varies significantly across different user experience factors. This variability reflects the different contexts in which remote wipe might be encountered.

4. App Protection Policies had the lowest standard deviation (0.49), indicating consistent moderate impact across user experience factors. Users experience similar levels of restriction across different applications.

IX. TECHNICAL IMPLEMENTATION

A. Microsoft Graph API Integration

The implementation leverages Microsoft Graph API for programmatic access to Intune features. This integration enables automated device management, policy application, and compliance monitoring. The Graph API provides RESTful endpoints for all Intune functionalities, allowing organizations to build custom management tools and automate routine tasks.

©2025 IEEE — All rights reserved.

B. Integration with Existing Systems

The implementation includes integration with:

- Active Directory / Azure AD: For user authentication and identity management
- Microsoft Defender for Endpoint: For threat detection and response
- WinPak Attendance System: For context-aware policy enforcement based on physical presence
- SIEM Systems: For security logging and alerting

C. Performance Optimization

Performance testing revealed that policy application times average under 5 minutes for 95% of devices. The system handles up to 10,000 concurrent devices with minimal latency. Optimization techniques include policy caching, incremental updates, and prioritized policy application for critical security settings.

D. Security Considerations

The implementation adheres to defense-in-depth principles:

- All API communications use TLS 1.2 or higher
- Certificates are rotated every 90 days
- Access tokens have a maximum lifetime of 60 minutes
- All actions are logged for audit purposes
- Regular penetration testing is conducted

X. FUTURE WORK

While this research provides valuable insights into Microsoft Intune's effectiveness, several areas warrant further investigation:

A. AI and Machine Learning Integration

Future research could explore the integration of AI/ML capabilities for predictive security. Machine learning models could analyze device behavior patterns to detect anomalies before they become security incidents. Microsoft's upcoming security AI features could be evaluated for their effectiveness in reducing false positives and improving threat detection.

B. Extended Platform Support

As new operating systems and device types emerge, research should evaluate Intune's support for:

- Linux workstations in development environments
- IoT devices and embedded systems
- AR/VR devices for training and collaboration
- Wearable technology in healthcare and manufacturing

C. Long-term User Experience Studies

Longitudinal studies could track user experience over time to determine:

- Whether users adapt to security measures and perceive less impact over time
- The optimal balance between security and productivity in different roles
- The effectiveness of user training in mitigating security feature impact

D. Cross-Platform Comparison

Comparative research could evaluate Intune against other MDM solutions:

- VMware Workspace ONE
- Jamf Pro for Apple devices
- IBM MaaS360
- Google Endpoint Management

E. Zero Trust Architecture Integration

As organizations adopt Zero Trust principles, research could explore:

- How Intune contributes to Zero Trust maturity
- Integration with other Zero Trust components
- Measurement of Zero Trust implementation effectiveness

F. Compliance Automation

Future work could examine automated compliance reporting:

- Real-time compliance dashboards
- Automated audit evidence collection
- Integration with regulatory frameworks (GDPR, HIPAA, ISO 27001)

XI. CONCLUSION

This research evaluated Microsoft Intune' effectiveness in securing enterprise devices while balancing security features with user experience. Through testing across 650 devices in multiple organizations, the study revealed that Intune provides comprehensive security capabilities including device encryption, multi-factor authentication, app protection policies, conditional access, data loss prevention, remote wipe, and compliance policies. The statistical analysis showed that while features like device encryption have minimal impact on user experience (mean = 0.57), others such as multi-factor authentication significantly affect usability (mean = 1.57). Remote wipe demonstrated the most

©2025 IEEE — All rights reserved.

variable impact (standard deviation = 1.07), while app protection policies showed consistent effects across all user experience factors.

Microsoft Intune proves to be an effective solution for modern device management, but successful implementation requires careful consideration of organizational priorities. The trade-off analysis framework developed in this research enables evidence-based decision-making, helping organizations balance security requirements with user experience to achieve optimal security posture without compromising productivity. Intune should be viewed as a critical component of a broader, layered security strategy rather than a complete solution.

XII. REFERENCES

[1] De los Reyes, G., Macwan, S., Chawla, D., & Serban, C. (2012). Securing the mobile enterprise with network-based security and cloud computing. 35th IEEE Sarnoff Symposium.
[2] Song, X., & Yang, C.-H. (2017). Mobile Device Management System Based on AOSP and SELinux. IEEE Second International Conference on Data Science in Cyberspace (DSC).
[3] Baltazar, S., Barreto, L., Amaral, A., & Pereira, T. (2020). Enterprise Information Systems (EIS) as a key driver towards improved mobility. 2020 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC).
[4] Hasan, B., Dmitriyev, V., Gomez, J. M., & Kurzhofer, J. (2014). A framework along with guidelines for designing secure mobile enterprise applications.

[5] Batool, H., & Masood, A. (2020). Enterprise Mobile Device Management Requirements and Features. IEEE INFOCOM 2020 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS).
[6] Yucel, S. (2017). Evaluating Enterprise Mobility Strategy. 2017 International Conference on Computational Science and Computational Intelligence (CSCI).
[7] Yamin, M. M., & Katt, B. (2019). Mobile device management (MDM) technologies, issues and challenges. Proceedings of the 3rd International Conference on Cryptography, Security and Privacy (pp. 223-227).
[8] Medudula, M. K., Sagar, M., & Gandhi, R. P. (2016). Mobile Device: Applications, Over the Top Services, Identity Protection and BYOD Policy. In Telecom Management in Emerging Economies (pp. 207-227). Springer

[9] Raj Kumar Koneru, Pattabhi Rama Rao Dasari, Prajakt Deshpande, Vivek Iyer, Rajendra Komandur, Aravind Perumal, Sriram Ramanathan, Matthew Terry, Vamsi Krishna Vagvala, Sathyanarayana Vennapusala, et al. (2016). Mobile application management systems and methods thereof (U.S. Patent No. 9,405,723). U.S. Patent and Trademark Office.

[10] Suduca, A. M., & Bizoa, M. (2022). AI shapes the future of web conferencing platforms. 9th International Conference on Information Technology and Quantitative Management.
[11] Liu, W., Wang, X., & Peng, W. (2019). Secure Remote Multi-Factor Authentication Scheme Based on Chaotic Map Zero-knowledge proof for Crowdsourcing Internet of Things. IEEE Access, 7, 1-1.

©2025 IEEE — All rights reserved.